

HOW LEADING SBCs STACK UP

Brand/SBC Model	Processor Platform	AI Acceleration	Form Factor/Type	Key Strength	Typical Applications
ADLINK SBC-35 MTL	Intel Core Ultra (x86)	Integrated NPU + GPU	3.5-inch (8.8cm) SBC	Balanced AI + industrial design	Robotics, vision AI, smart factories
ADLINK SBC-35 ASL	Intel Atom x7000RE	NPU/low-power AI	3.5-inch (8.8cm) SBC	Fanless, rugged, low power	Automation, Industrial IoT gateways, outdoor IoT
Nvidia Jetson AGX Orin	Nvidia Orin SoC (ARM)	High-end GPU + DLA	Dev SBC	Top AI perf	Autonomous systems, advanced robotics
Nvidia Jetson Orin Nano/ Super	Orin variants	GPU + AI accelerators	Dev SBC	Strong inference	Robotics, AI devices
Google Coral Dev Board	NXP i.MX + Edge TPU	Edge TPU (~4 TOPS+)	SBC	TensorFlow support	Edge ML & CV
ASUS Tinker Edge R / Tinker Board series	Rockchip RK3399Pro and variants	NPU (~3 TOPS)	SBC	Maker + AI	ML inference, vision
Radxa Cubie A7A	Allwinner A733	3-TOPS NPU	SBC	Versatile AI	Vision & compute
Orange Pi Alpro (20T)	Huawei Ascend + ARM cores	DaVinci NPU (8~20 TOPS)	SBC	High AI perf	Inference & edge AI
Banana Pi BPI-F3	SpacemiT K1 (RISC-V)	GPU-assist (~2 TOPS)	SBC	Emerging AI board	IoT + vision
BeagleBone AI-64	TI TDA4VM	Multiple DSP AI engines	SBC	Open source AI	Robotics, analytics
Raspberry Pi 5 + AI accel HAT	BCM2712 + Hailo / similar	External AI module (~13 TOPS)	SBC	Massive ecosystem	AI prototyping
PHYTEC phyBOARD-Pollux	NXP i.MX 8M Plus	Integrated NPU (~2.3 TOPS)	SBC	Industrial ARM AI	Machine vision, HMI
Particle Tachyon SBC	Qualcomm Snapdragon QCM6490	Integrated NPU (~12 TOPS)	Credit card SBC	5G + AI	Mobile + edge apps
Geniatech AI SBCs (various)	ARM/RISC-V/varied	Platform NPUs	SBC/embedded	Flexible industrial OEM	AIoT/automation
Vicharak Axon SBC	Rockchip RK3588	6-TOPS NPU + Mali GPU	SBC	Strong AI + 4K	Edge AI and multimedia
Libre Computer Alta (AML-A311D)	Amlogic A311D	~5 TOPS NPU	SBC	Open platform AI	AI inference and IoT
Libre Computer Solitude (S905D3)	Amlogic S905D3	~1.2 TOPS NPU	SBC	Value AI board	Lightweight ML
HiKey970 SBC	HiSilicon Kirin 970	Dedicated NPU platform	SBC	Early NPU board	DL/AI models
BeagleY-AI	Community AI-SBC	AI accelerators	SBC	Open/niche	Embedded AI
Avnet MaaXBoard OSM93	Snapdragon/NPU	Dedicated AI engine	SBC	Qualcomm edge AI	Low-power ML
MemryX MX3 SBC	MemryX AI engine	High-perf AI core	SBC	AI hardware focus	ML acceleration
AAEON PICO-MTU4	Intel Core Ultra	Integrated AI engines	Pico-ITX	High compute in tiny footprint	Embedded AI, compact HMIs
Advantech RSB-3810	MediaTek Genio 1200	On-chip AI (~5 TOPS)	Pico-ITX	Power-efficient edge AI	Retail, automation, smart devices
Kontron SBC-AML/ADN	Intel Atom/N-series	Integrated graphics/AI	3.5-inch (8.8cm)	Long lifecycle, industrial focus	Rail, automation, infrastructure

this modular thinking in practice. It enables I/O customisation, including USB, Ethernet, digital I/O, and camera interfaces, through a family of rugged, lockable modules that can be added without modifying the main processor SBC. The use of FPC-based cable options further supports flexible mechanical layouts, allowing vertical or parallel stacking. This is especially useful for space-constrained designs, as well as environments where high vibration is a

concern, where rigid connectors may introduce reliability risks.

Similarly, modular design philosophies underpin Advantech's RSB Pico-ITX family and AAEON's UP board family, both built around the carrier board paradigm. This model is particularly suited to industrial and robotics applications.

These modular SBC environments typically support Windows, Linux, Ubuntu, and Yocto, alongside AI/ML stacks, such as OpenVINO, TensorRT,

TensorFlow Lite, and PyTorch.

For OEMs, the implications are profound: modular expansion means faster time-to-market, lower certification and redesign costs, and the potential to reuse a common hardware platform across multiple products and industries. As product life cycles shorten while demand for customisation increases, modular SBC architectures are becoming a strategic enabler rather than a convenience feature.